

KANDULA JITHENDRA SUBRAMANYAM

M.Tech AI & Data Science | Agentic AI | Multi-Agent Systems | Quantitative Finance | Full-Stack Dev

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PROFESSIONAL SUMMARY

M.Tech Artificial Intelligence and Data Science candidate with hands-on experience in agentic AI, multi-agent systems, portfolio governance, quantitative finance, explainable AI, and full-stack development. Built an Agentic AI Portfolio Governance Chatbot and Multi-Agent Adaptive Portfolio Governance System using specialized agents for planning, data retrieval, technical analysis, market-regime detection, instability analysis, portfolio optimization, governance validation, explainability, and verification. Research manuscripts submitted to Elsevier EAAI and Springer conference proceedings are currently under review. Skilled in Python, LangChain/LangGraph, CVXPY, NetworkX, MongoDB, React, Next.js, Flask, REST APIs, financial risk analytics, and compliance-aware AI systems.

TECHNICAL SKILLS

AI, ML & Research	Artificial Intelligence, Machine Learning, Deep Learning, Agentic AI, Multi-Agent Systems, Explainable AI, Retrieval-Augmented Generation, Prompt Engineering
Finance & Optimization	Quantitative Finance, Portfolio Optimization, CVaR, Drawdown, Volatility, RSI, MACD, Bollinger Bands, Market Regime Detection, Compliance-Aware AI
Programming & Data	Python, NumPy, Pandas, SciPy, Scikit-learn, SQL, MongoDB, Git, GitHub
Agentic AI & LLM Tools	LangChain, LangGraph, Ollama, Mistral-7B, Agent Memory, Verification Frameworks, Audit Logging, Source Grounding
Optimization & Graph Tools	CVXPY, CLARABEL, NetworkX, Risk-Aware Portfolio Construction, Graph-Based Reasoning
Full-Stack Development	React, Next.js, Flask, REST APIs, TypeScript, JavaScript, HTML, CSS, Gradio, AQICN API
Embedded / IoT	Embedded Systems, IoT Sensors, Robotics, Image Processing Concepts

PROFESSIONAL EXPERIENCE

Thesis Researcher – Agentic AI Portfolio Governance

Oct 2025 – Apr 2026

K J Somaiya College of Engineering | Mumbai

- Developed an Agentic AI Portfolio Governance Chatbot for stock portfolio management using a modular multi-agent architecture.
- Designed specialized agents for task planning, data retrieval, technical analysis, market-regime detection, instability analysis, portfolio optimization, governance validation, explainability, self-verification, and final response generation.
- Implemented audit logging, confidence scoring, evidence tracking, source grounding, anti-hallucination controls, and compliance-aware decision support to improve reliability of financial AI outputs.
- Built technical-analysis modules for RSI, MACD, Bollinger Bands, support/resistance, trend analysis, volatility, drawdown, and Bloomberg-style reporting.
- Integrated Python, LangChain/LangGraph, CVXPY, CLARABEL, NetworkX, MongoDB, Gradio, Ollama, Mistral-7B, YFinance, NumPy, Pandas, SciPy, and financial datasets.

Full-Stack Developer Intern

May 2025 – Aug 2025

ScholarRankAI | Remote

- Designed and delivered responsive React-based user interfaces for production workflows.
- Integrated third-party REST APIs and supported end-to-end frontend functionality.
- Participated in Agile stand-ups, sprint planning, debugging cycles, code reviews, and performance optimization.
- Reduced page-load bottlenecks through targeted frontend improvements and enhanced usability across application screens.

UI/UX Developer Intern

Mar 2024 – May 2024

MNJ Software Pvt. Ltd. | India

- Translated UI/UX wireframes into production-ready HTML, CSS, and JavaScript interfaces.
- Maintained visual consistency across screens and collaborated with backend engineers on frontend-backend integration.
- Improved page-load speed by 25% and increased user engagement metrics by 15% through targeted UI and performance enhancements.

PROJECTS

Agentic AI Portfolio Governance Chatbot

Oct 2025 – Apr 2026

- Built a personal chatbot for stock portfolio management using a multi-agent AI framework with planning, data retrieval, technical analysis, regime detection, instability analysis, portfolio optimization, governance validation, explainability, verification, and response generation agents.
- Implemented audit logs, confidence scores, evidence tracking, source grounding, anti-hallucination controls, and compliance-aware outputs.
- Stack: Python, LangChain/LangGraph, CVXPY, CLARABEL, NetworkX, MongoDB, Gradio, Ollama, Mistral-7B, YFinance, NumPy, Pandas, SciPy.

Multi-Agent Adaptive Portfolio Governance System

Jan 2026 – Apr 2026

- Designed a financial decision-intelligence system combining agentic orchestration, risk-aware portfolio construction, regime-aware adaptation, instability detection, optimization, explainability, governance checks, and response verification.
- Built the system to support safer financial reasoning through evidence-backed responses, governance approval checks, transparent explanations, and audit-ready outputs.

Personalised AQI – Global Air Quality Forecasting System

Feb 2023 – Mar 2025

- Built a full-stack AQI forecasting platform using live AQICN.org data to predict city-level air quality and pollutant trends worldwide.
- Implemented ML models for AQI, PM2.5, O3, and NO2 forecasting with explainable health-risk insights.
- Designed a Next.js dashboard for real-time and predicted AQI trend visualization across cities, states, and countries.
- Stack: Next.js, Python, Flask, MongoDB, Machine Learning, Explainable AI, AQICN API.

Swarm Robots for Agriculture

Aug 2022 – Jul 2023

- Built a swarm robotics agriculture project using embedded systems, IoT sensors, autonomous coordination, and image-processing concepts.
- Worked on plant disease detection, sensor-based monitoring, robot coordination, and agriculture-focused automation.
- Combined electronics, communication systems, computer vision concepts, and embedded implementation for agricultural assistance.

RESEARCH PAPERS / PUBLICATIONS

Multi-Agent Adaptive Portfolio Governance System

Under Review, 2026

Elsevier – Engineering Applications of Artificial Intelligence

Focus: agentic AI orchestration, portfolio optimization, explainable governance, market-regime adaptation, instability detection, anti-hallucination safeguards, auditability, and transparent financial decision intelligence.

Focus: agentic AI, multi-agent portfolio governance, explainable financial intelligence, compliance-aware portfolio decision support, and audit-ready AI workflows.

EDUCATION

M.Tech – Artificial Intelligence and Data Science

Aug 2024 – Apr 2026

K J Somaiya College of Engineering | Mumbai | CGPA: 8.06

Thesis: Personal Chatbot for Stock Portfolio Management with Agentic AI Framework. Research focus: agentic AI, multiagent portfolio governance, explainable AI, quantitative finance, portfolio optimization, market-regime analysis, riskaware portfolio construction, governance validation, and compliance-aware financial intelligence systems.

B.Tech – Electronics and Communication Engineering

Aug 2019 – Jul 2023

Presidency University | Bangalore | CGPA: 7.77

Final-year project: Swarm Robots for Agriculture using embedded systems, image processing, IoT sensors, autonomous coordination, and plant disease detection.

Intermediate, MPC – Sri Chaitanya Junior College

Jun 2017 – Jun 2019

CGPA: 9.28

10th Grade – T V N R M National High School

Jun 2016 – Jun 2017

CGPA: 9.5

CERTIFICATIONS & TEST SCORES

Certifications: Smart Manager Certificate (CII), Python Programming, Machine Learning, C Programming

Standardized Tests: GATE Computer Science: 23.2 | GATE AI & Data Science: 33.3 | JEE Mains: 85.6 Percentile

ACHIEVEMENTS & HIGHLIGHTS

- Submitted research manuscripts to Elsevier Engineering Applications of Artificial Intelligence and Springer conference proceedings (both under review, 2026).
- Developed an end-to-end agentic AI portfolio governance chatbot with multi-agent orchestration, optimization, explainability, verification, and compliance-aware decision support.
- Improved UI performance by 25% and user engagement by 15% during UI/UX internship at MNJ Software.
- Built AQI forecasting platform using live AQICN data, ML forecasting, explainable insights, and Next.js dashboard.
- Built agriculture-focused swarm robotics project combining embedded systems, IoT sensors, image-processing, and autonomous coordination.

LANGUAGES

English: Fluent | Telugu: Native | Hindi: Intermediate